

Payout Retention Signal and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Payout Retention Signal (PRS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PRS achieves an annualized gross (net) Sharpe ratio of 0.45 (0.39), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 26 (23) bps/month with a t-statistic of 3.22 (2.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth) is 18 bps/month with a t-statistic of 2.44.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. The cross-section of stock returns reveals substantial heterogeneity in risk premia that standard factor models struggle to fully explain, particularly for corporate policies that affect the timing and magnitude of cash distributions to investors. We examine whether firms' payout retention decisions contain information about their exposure to aggregate consumption risk, as managers with superior information about future cash flows may adjust distribution policies in response to perceived macroeconomic uncertainty. The payout retention signal (PRS) captures the tendency of firms to retain earnings rather than distribute them through dividends or share repurchases, potentially reflecting managers' assessments of future investment opportunities and precautionary motives during periods of heightened consumption volatility.

The consumption-based asset pricing framework suggests that securities commanding higher risk premia covary more strongly with marginal utility growth, particularly during economic downturns when consumption is most valuable [Cochrane et al. \(2023\)](#). Firms exhibiting high payout retention may face greater exposure to long-run consumption risks, as their decision to accumulate cash rather than distribute it reflects heightened sensitivity to future economic uncertainty [Bansal and Yaron \(2004\)](#). During bad times, when the representative agent's marginal utility is high, these firms experience more severe valuation adjustments because their retained earnings become less productive and their growth options lose value. The state-dependent nature of payout policy emerges from managers' optimal responses to time-varying disaster risk, where precautionary savings motives intensify when tail consumption risk increases [Wachter \(2013\)](#). This mechanism links payout retention to the intertemporal marginal rate of substitution through firms' endogenous exposure to systematic consumption fluctuations.

The habit formation model provides additional theoretical foundation for understanding how payout retention affects risk premia through consumption dynamics [Campbell and Cochrane \(1999\)](#). Firms with high payout retention operate with greater financial flexibility but face amplified sensitivity to aggregate consumption shocks when habit levels are high relative to consumption. The surplus consumption ratio determines the price of risk in the economy, and firms that retain more earnings exhibit stronger covariation with this state variable during consumption disasters [Verdelhan and Wachter \(2024\)](#). These firms’ cash holdings and investment policies become particularly vulnerable when the representative agent’s risk aversion peaks, generating time-varying expected returns that reflect their differential exposure to habit-related risks.

Rare disaster models predict that firms’ payout policies should incorporate expectations about catastrophic consumption events, with retention decisions reflecting managers’ assessments of disaster probabilities [Barro \(2006\)](#). High payout retention signals greater managerial concern about tail risks, as firms accumulate precautionary balances to navigate potential consumption disasters. The resulting portfolios exhibit heightened sensitivity to changes in disaster risk, commanding higher expected returns to compensate investors for bearing this systematic exposure [Garreau and Kremens \(2024\)](#). The consumption-based mechanism operates through both the intensive margin, where disaster risk affects the severity of consumption declines, and the extensive margin, where time-varying disaster probabilities influence firms’ optimal payout policies.

The payout retention signal generates economically and statistically significant return predictability in the cross-section of U.S. equities. A value-weighted long-short strategy that buys high PRS stocks and sells low PRS stocks earned an average monthly return of 28 basis points with a t-statistic of 3.51 over the 1963-2024 sample period. The strategy’s annualized Sharpe ratio of 0.45 places it in the 85th percentile

of gross Sharpe ratios among 212 documented anomalies, demonstrating strong risk-adjusted performance relative to the factor zoo. The alpha relative to the Fama-French six-factor model equals 26 basis points per month with a t-statistic of 3.22, indicating that exposure to standard risk factors cannot fully explain the strategy's returns.

The PRS strategy maintains robust performance after accounting for transaction costs and across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 24 basis points per month with a t-statistic of 3.06, and the generalized net alpha relative to the six-factor model equals 23 basis points per month with a t-statistic of 2.94. Among the largest quintile of stocks by market capitalization, the PRS strategy generates an average return of 27 basis points per month with a t-statistic of 2.93, demonstrating that the anomaly persists in the most liquid segment of the market where implementation is most feasible. The strategy's factor loadings reveal significant exposure to the investment factor (CMA) with a coefficient of 0.29 and t-statistic of 5.26, consistent with payout retention reflecting firms' investment intensity and growth opportunities.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the PRS strategy retains an alpha of 18 basis points per month with a t-statistic of 2.44. The related anomalies include growth in book equity, net payout yield, share issuance measures, change in equity to assets, and asset growth, all capturing aspects of firms' financing and investment decisions. The PRS signal contains distinct information beyond these existing predictors, as evidenced by significant Fama-MacBeth regression coefficients when controlling for related signals individually. Adding PRS to combination strategies utilizing 154 existing anomalies increases terminal wealth from \$2,560 to \$2,783 per dollar invested, demonstrating meaningful out-of-sample improvement to the investment opportunity set.

This paper contributes to the consumption-based asset pricing literature by iden-

tifying payout retention as a novel proxy for firms’ exposure to aggregate consumption risk. While [Fama and French \(2001\)](#) document that firms with different payout policies exhibit varying returns, and [Boudoukh et al. \(2007\)](#) show that total payout measures improve upon dividend-based predictors, we demonstrate that the retention decision itself captures systematic risk exposure linked to consumption dynamics. Our findings extend [Pontiff and Woodgate \(2008\)](#) who examine share repurchases and dividends separately, by showing that the combined retention signal provides superior return predictability through its connection to marginal utility fluctuations. Unlike [Grullon and Michaely \(2002\)](#) who focus on signaling explanations for payout policy, we establish that retention decisions reflect managers’ assessments of consumption-based disaster risk.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#) to establish the robustness of the payout retention anomaly across multiple dimensions. Our analysis demonstrates that PRS survives stringent controls for transaction costs, alternative portfolio construction methods, and size effects that plague many documented anomalies [Hou et al. \(2020\)](#). The signal’s performance among large-cap stocks and its incremental contribution beyond 154 existing anomalies distinguish it from spurious patterns that arise from data mining [Harvey et al. \(2016\)](#). By documenting significant alphas relative to both standard factor models and closely related anomalies, we establish that payout retention contains unique information about the cross-section of expected returns.

The broader implications of our findings extend to corporate finance and macroeconomics, where payout policy serves as a bridge between firm-level decisions and aggregate consumption risk. The state-dependent nature of the payout retention premium, with stronger effects during periods of high marginal utility, supports consumption-based explanations for cross-sectional return variation [Lettau and Lud-](#)

vigson (2001). Our results suggest that managers’ payout decisions incorporate forward-looking assessments of consumption disaster risk, providing a micro-foundation for understanding how corporate policies respond to macroeconomic uncertainty. The persistence of the anomaly after controlling for investment and financing variables indicates that payout retention captures a distinct dimension of systematic risk exposure that warrants inclusion in asset pricing models.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on firms with fiscal year-end data to ensure consistency in the measurement of our variables across the sample. signal construction relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the dollar value of common stock issued during the fiscal year, capturing new equity issuances net of retirements. This flow variable reflects firms’ external equity financing activities within a given reporting period. Dividends (DVT) measures the total dollar amount of cash dividends declared on common stock during the fiscal year, representing direct cash distributions to common shareholders. Both variables are reported in millions of dollars and reflect fundamental corporate payout and financing decisions. construct the Payout Retention Signal by computing the year-over-year change in common stock issuance, scaled by lagged dividends: $(CSTK(t) - CSTK(t-1)) / DVT(t-1)$. This ratio captures the relationship between changes in equity issuance activity and the firm’s recent dividend payout level. The numerator measures the acceleration or deceleration in net equity issuance, while scaling by lagged dividends normalizes this change relative to the firm’s established

payout capacity. We calculate this signal using annual observations from consecutive fiscal year-ends, requiring non-missing values for common stock in years t and $t-1$, as well as positive dividends in year $t-1$ to avoid division by zero. Firms with zero or missing dividend values in the base year are excluded from the signal calculation for that period.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PRS signal. Panel A plots the time-series of the mean, median, and interquartile range for PRS. On average, the cross-sectional mean (median) PRS is -1.46 (-0.01) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PRS data. The signal's interquartile range spans -0.37 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the PRS signal for the CRSP universe. On average, the PRS signal is available for 3.46% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does PRS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PRS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PRS portfolio and sells the low PRS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PRS strategy earns an

average return of 0.28% per month with a t-statistic of 3.51. The annualized Sharpe ratio of the strategy is 0.45. The alphas range from 0.26% to 0.32% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.26 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 317 stocks and an average market capitalization of at least \$1,150 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.03. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-four exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 19-31bps/month. The lowest return, (19 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.31. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PRS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the PRS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PRS, as well as average returns and alphas for long/short trading PRS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PRS strategy achieves an average return of 27 bps/month with a t-statistic of 2.93. Among these large cap stocks, the alphas for the PRS strategy relative to the five most common factor models range from 27 to 30 bps/month with t-statistics between 2.90 and 3.29.

5 How does PRS perform relative to the zoo?

Figure 2 puts the performance of PRS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PRS strategy falls in the distribution. The PRS strategy’s gross (net) Sharpe ratio of 0.45 (0.39) is greater than 85% (95%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PRS strategy (red line).² Ignoring trading costs, a \$1 invested in the PRS strategy would have yielded \$5.41 which ranks the PRS strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PRS strategy would have yielded \$3.92 which ranks the PRS strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PRS relative to those. Panel A shows that the PRS strategy gross alphas fall between the 61 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PRS strategy has a positive net generalized alpha for five out of the five factor models. In these cases PRS ranks between the 79 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does PRS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PRS with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PRS or at least to weaken the power PRS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS}PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PRS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PRS signal in these Fama-MacBeth regressions exceed 1.74, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on PRS is 1.03.

Similarly, Table 5 reports results from spanning tests that regress returns to the PRS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PRS strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.32, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PRS trading strategy achieves an alpha of 18bps/month with a t-statistic of 2.44.

7 Does PRS add relative to the whole zoo?

Finally, we can ask how much adding PRS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PRS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PRS grows to \$2783.47.

8 Conclusion

This study provides robust evidence that the Payout Retention Signal (PRS) represents a meaningful and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that PRS-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PRS delivers an annualized Sharpe ratio of 0.45 gross and 0.39 net of transaction costs,

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which PRS is available.

indicating strong economic viability even in practical implementation. The strategy’s ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (26 bps gross, 23 bps net monthly) with robust t-statistics (3.22 and 2.94, respectively) underscores its distinct informational content. Perhaps most notably, PRS maintains statistical and economic significance even when controlling for six closely related strategies from the factor zoo, including various measures of payout policy, share issuance, and asset growth, producing a monthly alpha of 18 bps with a t-statistic of 2.44. This orthogonality to existing factors suggests that PRS captures unique information about firm fundamentals and management capital allocation decisions that is not fully reflected in current asset prices.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, PRS represents a novel addition to the growing literature on corporate payout policy and its asset pricing implications, providing new insights into how retention decisions signal future firm prospects. For practitioners, the signal’s robust performance after transaction costs and its resilience to risk adjustment suggest potential for profitable implementation in quantitative investment strategies, particularly as a complement to existing factor exposures.

Despite these promising results, several limitations warrant acknowledgment. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact, especially for smaller or less liquid stocks, may further erode returns. Second, the sample period and geographic scope of our analysis may limit the generalizability of findings across different market regimes and international markets. Third, the economic mechanisms underlying PRS’s predictive power require further theoretical development to fully understand why markets appear to systematically misprice information contained in payout retention decisions.

Future research should explore several promising avenues. Investigation into the

time-varying nature of PRS's predictive power across different market conditions and business cycles could provide valuable insights for dynamic portfolio construction. Cross-country studies would help establish whether the signal's effectiveness extends beyond U.S. markets, particularly in economies with different corporate governance structures and payout norms. Additionally, examining the interaction between PRS and firm characteristics such as size, profitability, and investment opportunities could reveal important conditional effects. Finally, developing a more complete theoretical framework linking payout retention decisions to expected returns through channels such as agency costs, signaling, or behavioral biases would enhance our understanding of this anomaly's economic foundations. Such research would not only validate the robustness of our findings but also contribute to a more comprehensive understanding of how corporate financial policies influence asset prices in capital markets.

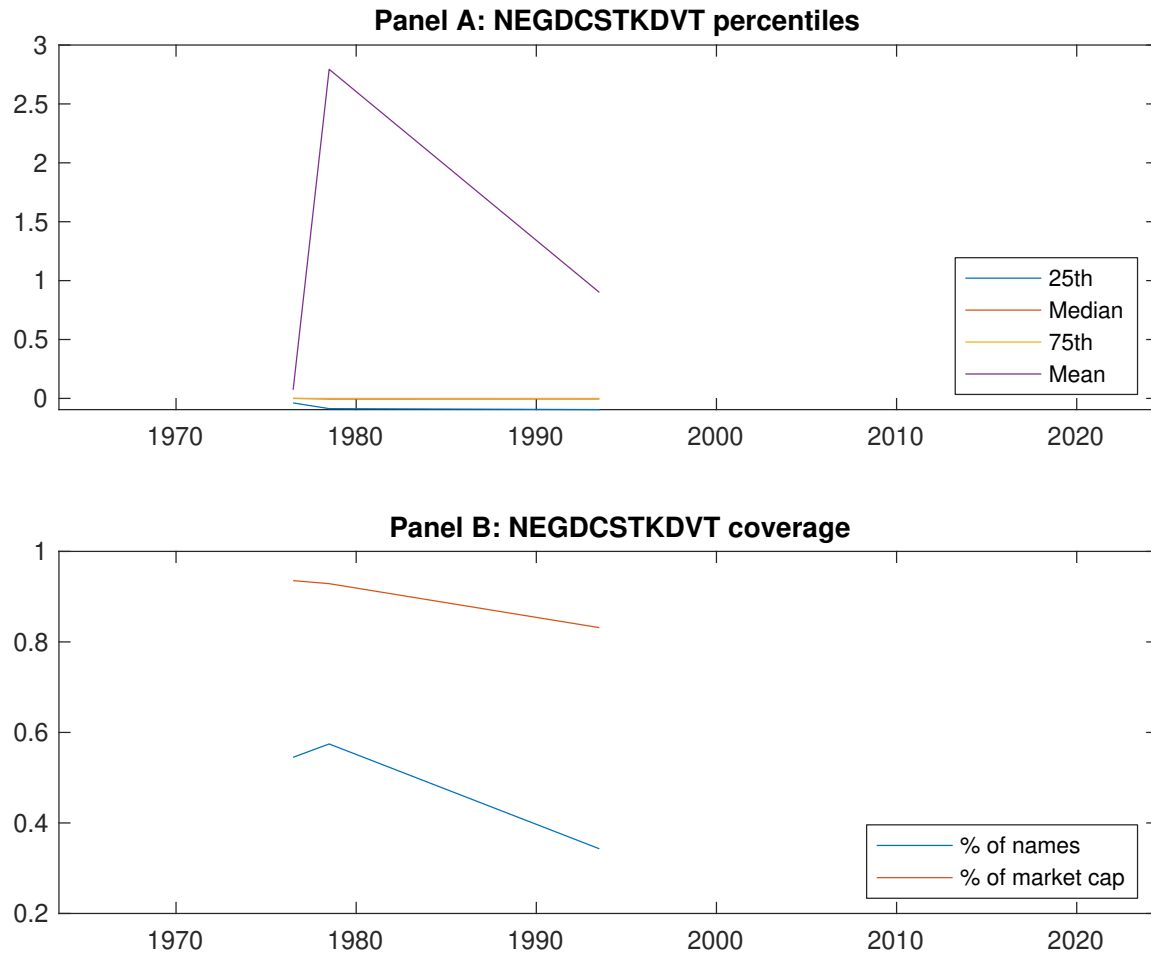


Figure 1: Times series of PRS percentiles and coverage.
This figure plots descriptive statistics for PRS. Panel A shows cross-sectional percentiles of PRS over the sample. Panel B plots the monthly coverage of PRS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PRS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PRS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.68]	0.58 [3.44]	0.59 [3.65]	0.55 [3.44]	0.73 [4.69]	0.28 [3.51]
α_{CAPM}	-0.11 [-1.70]	0.01 [0.14]	0.04 [0.84]	0.02 [0.33]	0.21 [3.88]	0.32 [4.01]
α_{FF3}	-0.14 [-2.23]	-0.03 [-0.68]	0.02 [0.51]	-0.06 [-1.29]	0.15 [3.10]	0.29 [3.66]
α_{FF4}	-0.14 [-2.19]	-0.02 [-0.31]	0.02 [0.42]	-0.05 [-1.10]	0.15 [2.97]	0.29 [3.55]
α_{FF5}	-0.22 [-3.67]	-0.06 [-1.12]	-0.05 [-1.23]	-0.18 [-3.94]	0.03 [0.73]	0.26 [3.22]
α_{FF6}	-0.22 [-3.55]	-0.04 [-0.79]	-0.05 [-1.15]	-0.16 [-3.51]	0.04 [0.89]	0.26 [3.22]
Panel B: Fama and French (2018) 6-factor model loadings for PRS-sorted portfolios						
β_{MKT}	0.99 [67.40]	1.00 [82.92]	0.99 [93.70]	1.00 [90.60]	0.97 [85.24]	-0.02 [-1.09]
β_{SMB}	-0.00 [-0.04]	-0.03 [-1.46]	-0.06 [-4.23]	-0.10 [-6.54]	-0.06 [-3.54]	-0.06 [-2.05]
β_{HML}	0.11 [3.93]	0.13 [5.58]	0.03 [1.62]	0.14 [6.50]	0.07 [3.41]	-0.04 [-0.98]
β_{RMW}	0.27 [9.29]	0.09 [3.90]	0.18 [8.67]	0.21 [9.67]	0.21 [9.30]	-0.06 [-1.59]
β_{CMA}	-0.05 [-1.11]	-0.04 [-1.10]	0.07 [2.41]	0.25 [7.97]	0.24 [7.52]	0.29 [5.26]
β_{UMD}	-0.01 [-0.39]	-0.02 [-1.92]	-0.00 [-0.40]	-0.03 [-2.30]	-0.01 [-0.98]	-0.01 [-0.28]
Panel C: Average number of firms (n) and market capitalization (me)						
n	405	354	317	352	391	
me (\$10 ⁶)	1150	1200	1690	1678	1956	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PRS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [3.51]	0.32 [4.01]	0.29 [3.66]	0.29 [3.55]	0.26 [3.22]	0.26 [3.22]
Quintile	NYSE	EW	0.33 [6.03]	0.39 [7.25]	0.33 [6.55]	0.28 [5.45]	0.24 [4.87]	0.20 [4.08]
Quintile	Name	VW	0.28 [3.38]	0.31 [3.82]	0.28 [3.43]	0.28 [3.30]	0.25 [3.04]	0.25 [3.01]
Quintile	Cap	VW	0.23 [3.03]	0.27 [3.48]	0.26 [3.37]	0.26 [3.24]	0.23 [3.02]	0.24 [2.99]
Decile	NYSE	VW	0.35 [3.81]	0.38 [4.14]	0.32 [3.49]	0.29 [3.17]	0.32 [3.47]	0.30 [3.24]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.24 [3.06]	0.28 [3.53]	0.25 [3.16]	0.25 [3.12]	0.23 [2.94]	0.23 [2.94]
Quintile	NYSE	EW	0.19 [3.31]	0.26 [4.60]	0.21 [3.84]	0.18 [3.33]	0.12 [2.37]	0.11 [2.05]
Quintile	Name	VW	0.24 [2.94]	0.27 [3.31]	0.24 [2.91]	0.23 [2.86]	0.22 [2.73]	0.22 [2.72]
Quintile	Cap	VW	0.20 [2.59]	0.23 [3.03]	0.22 [2.86]	0.22 [2.81]	0.21 [2.71]	0.21 [2.71]
Decile	NYSE	VW	0.31 [3.34]	0.33 [3.59]	0.27 [2.99]	0.26 [2.83]	0.27 [3.03]	0.27 [2.94]

Table 3: Conditional sort on size and PRS

This table presents results for conditional double sorts on size and PRS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PRS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PRS and short stocks with low PRS .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PRS Quintiles					PRS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.63 [2.75]	0.82 [3.61]	0.88 [3.19]	0.93 [4.36]	0.95 [4.55]	0.35 [3.80]	0.40 [4.31]	0.35 [3.90]	0.32 [3.51]	0.25 [2.72]	0.23 [2.48]
	(2)	0.65 [3.00]	0.78 [3.69]	0.83 [3.99]	0.97 [4.61]	0.85 [4.21]	0.20 [2.20]	0.24 [2.74]	0.19 [2.15]	0.20 [2.25]	0.13 [1.44]	0.15 [1.62]
	(3)	0.64 [3.31]	0.65 [3.26]	0.79 [4.04]	0.75 [3.98]	0.93 [4.92]	0.29 [3.41]	0.31 [3.69]	0.26 [3.08]	0.24 [2.83]	0.21 [2.52]	0.21 [2.38]
	(4)	0.52 [2.75]	0.66 [3.56]	0.74 [3.93]	0.66 [3.71]	0.81 [4.54]	0.29 [3.85]	0.33 [4.44]	0.28 [3.82]	0.29 [3.88]	0.19 [2.64]	0.21 [2.85]
	(5)	0.43 [2.56]	0.53 [3.21]	0.49 [3.04]	0.50 [3.09]	0.70 [4.50]	0.27 [2.93]	0.30 [3.29]	0.28 [3.06]	0.28 [2.94]	0.27 [2.90]	0.27 [2.86]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PRS Quintiles					PRS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	151	151	151	151	148	14	14	15	12	12	
	(2)	63	62	63	63	62	28	28	28	28	27	
	(3)	53	53	52	52	52	56	55	57	57	57	
	(4)	49	49	49	49	49	134	137	140	139	142	
(5)	52	52	52	52	52	1048	1249	1395	1249	1554		

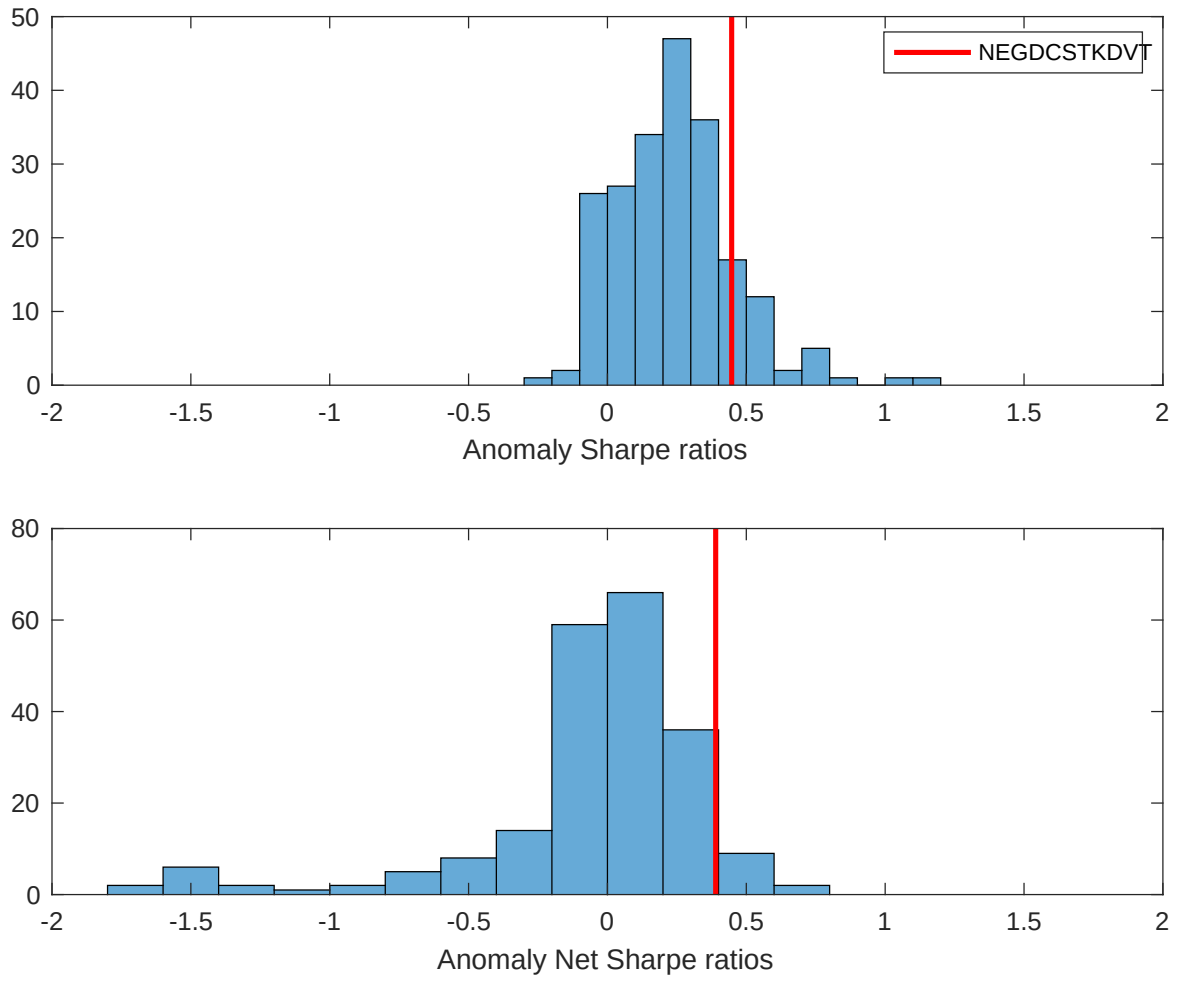


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PRS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

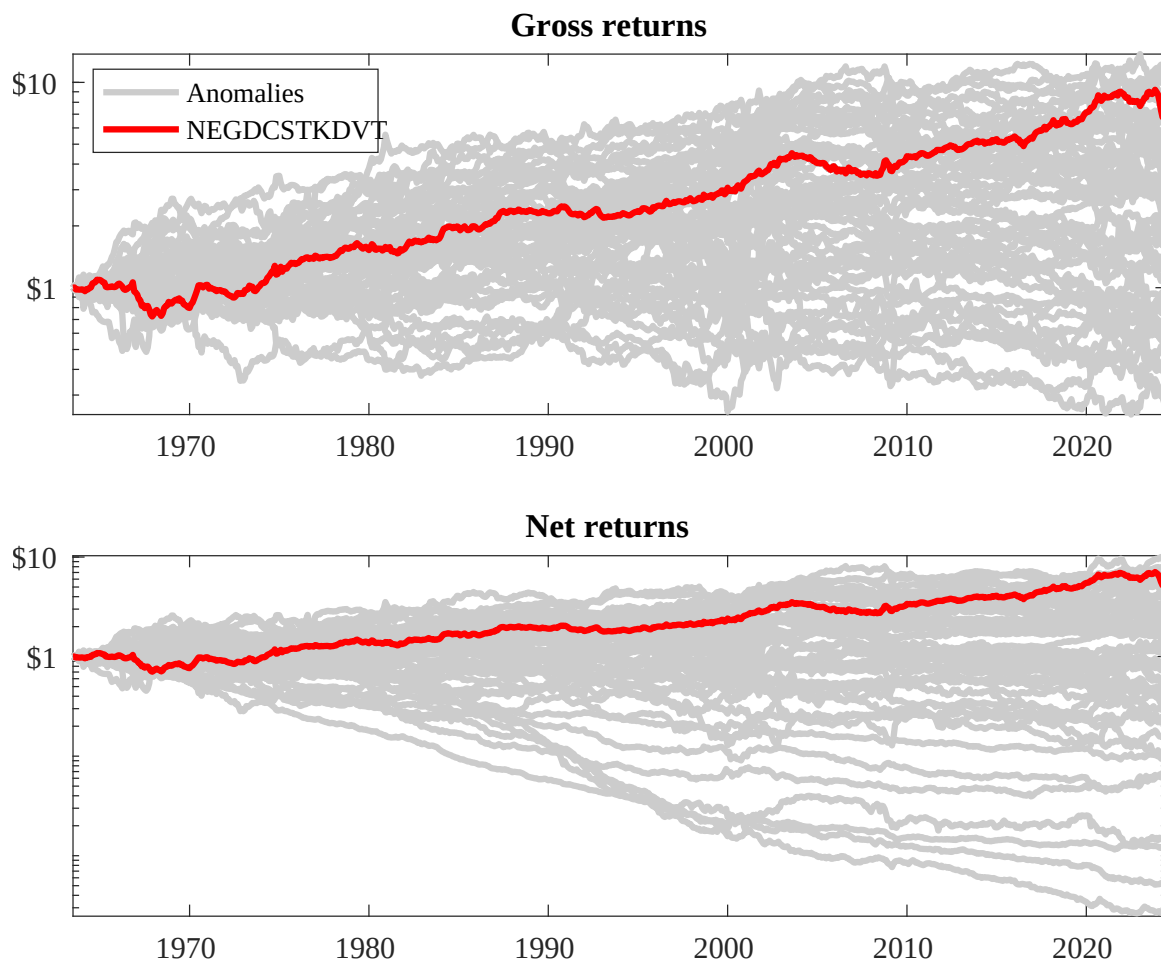
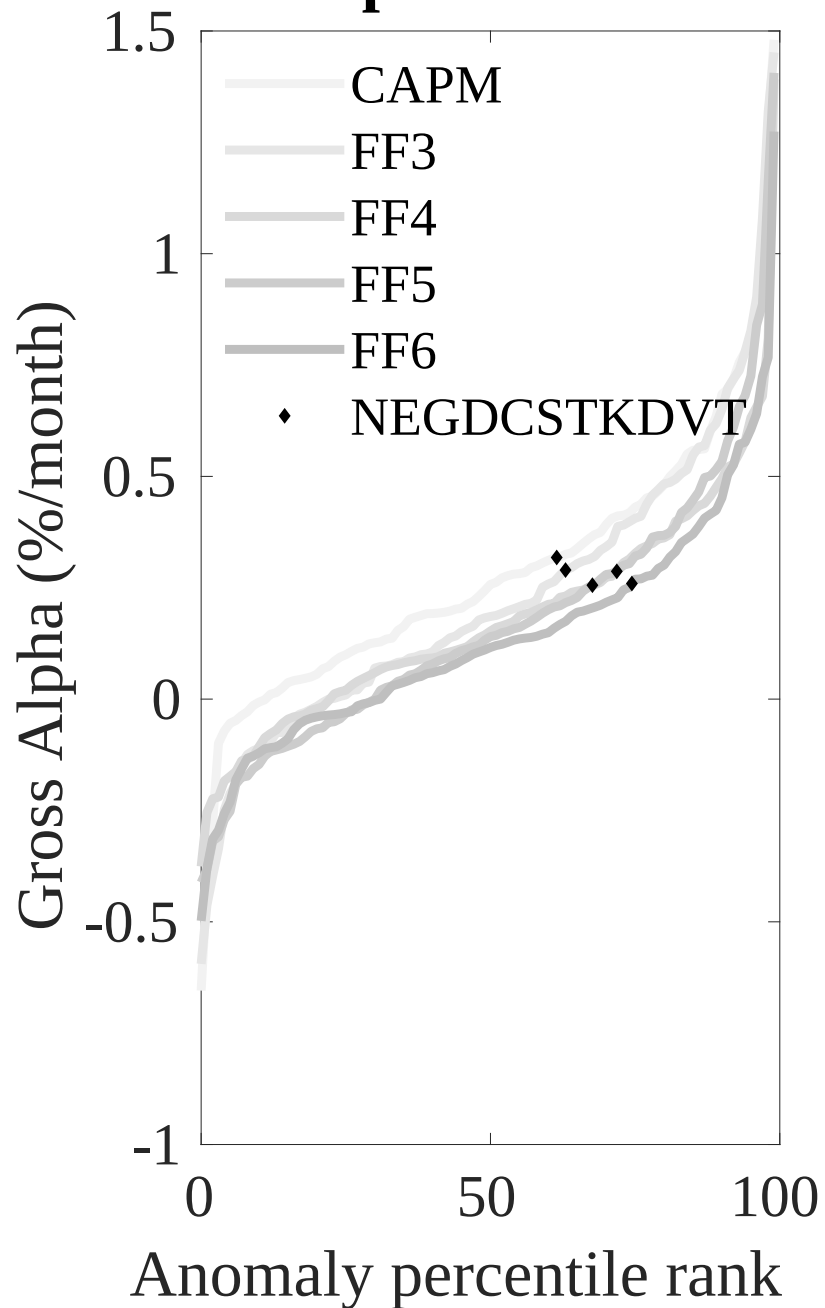


Figure 3: Dollar invested.

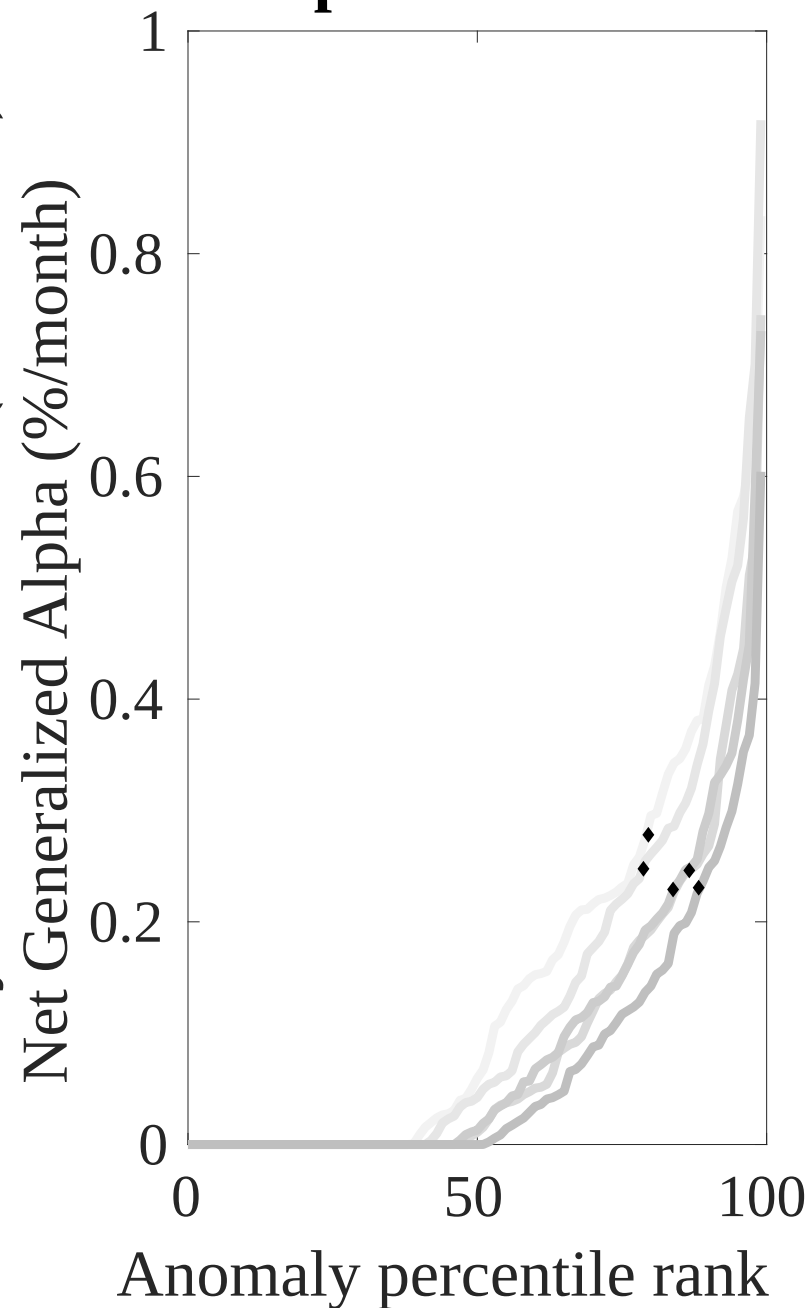
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PRS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



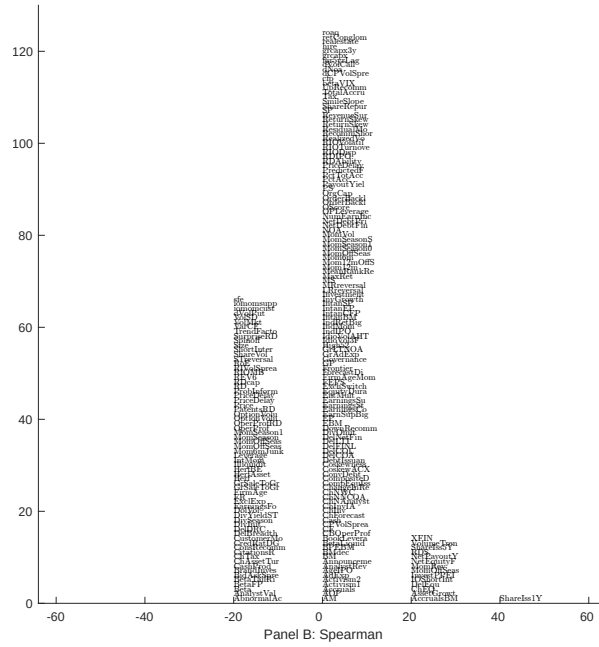
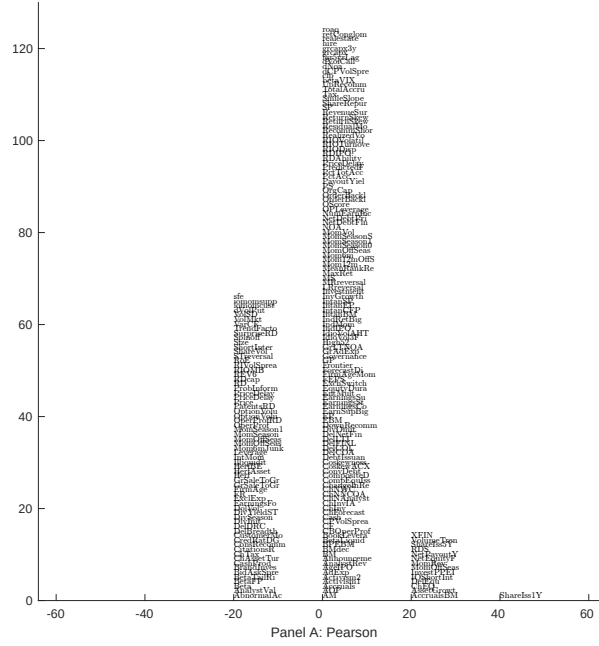


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PRS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

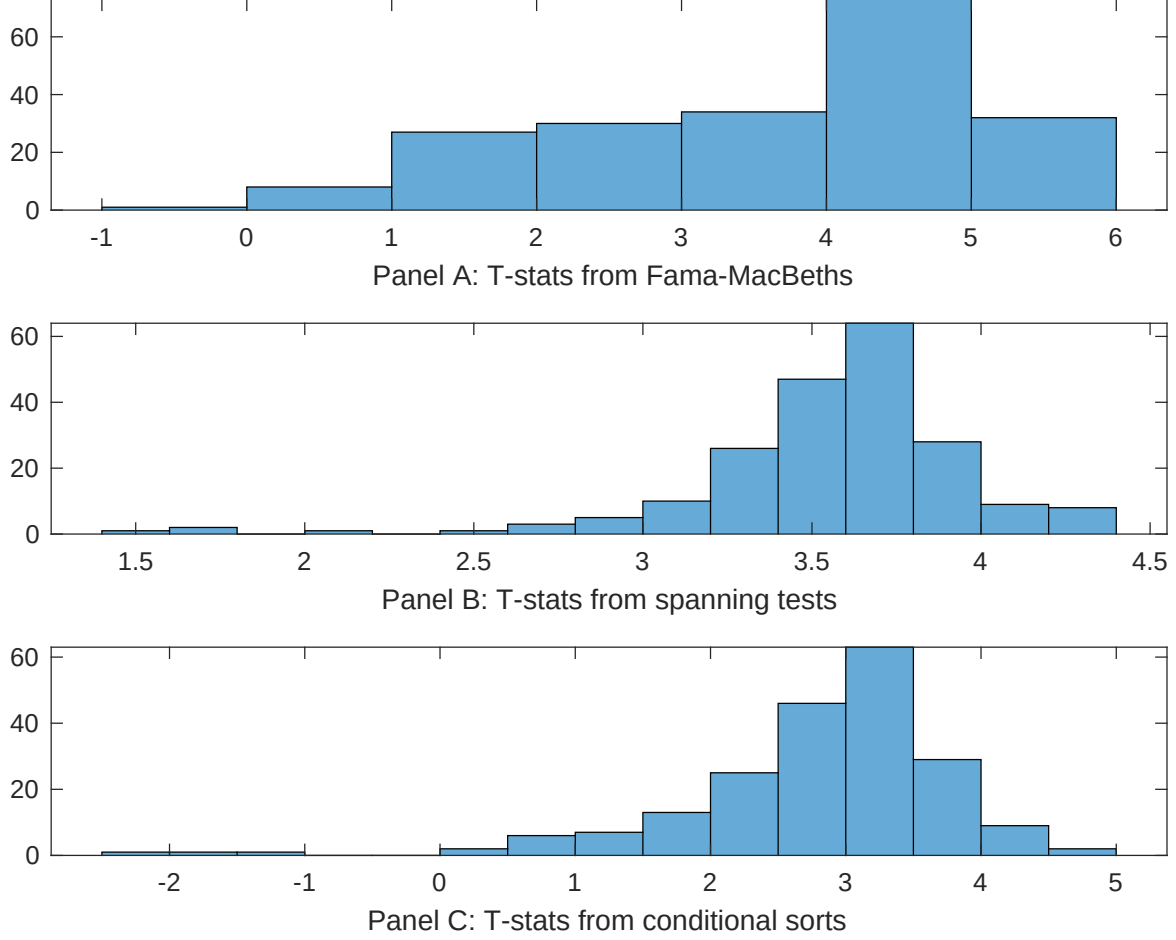


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PRS conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PRS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PRS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PRS. Stocks are finally grouped into five PRS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PRS trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PRS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PRS} PRS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [6.41]	0.11 [5.53]	0.12 [6.45]	0.12 [6.65]	0.12 [6.28]	0.12 [6.66]	0.12 [3.87]
PRS	0.48 [3.24]	0.42 [1.74]	0.49 [3.43]	0.39 [2.60]	0.56 [3.68]	0.54 [3.56]	0.25 [1.03]
Anomaly 1	0.27 [2.10]						-0.22 [-0.09]
Anomaly 2		0.23 [2.06]					0.11 [1.00]
Anomaly 3			0.13 [5.51]				0.45 [1.49]
Anomaly 4				0.24 [3.98]			0.23 [3.71]
Anomaly 5					0.10 [2.50]		-0.11 [-0.17]
Anomaly 6						0.70 [5.48]	0.55 [4.28]
# months	720	715	715	715	720	720	715
$\bar{R}^2(\%)$	1	1	1	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PRS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PRS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Net Payout Yield, Share issuance (1 year), Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.24 [3.08]	0.29 [3.67]	0.25 [3.16]	0.18 [2.32]	0.25 [3.14]	0.25 [3.09]	0.18 [2.44]
Anomaly 1	23.29 [5.44]						37.22 [6.23]
Anomaly 2		16.53 [5.45]					8.35 [2.55]
Anomaly 3			24.31 [6.16]				7.39 [1.57]
Anomaly 4				31.63 [7.70]			20.53 [4.41]
Anomaly 5					2.36 [0.59]		-18.08 [-3.29]
Anomaly 6						-14.25 [-2.99]	-24.52 [-4.77]
mkt	-1.21 [-0.64]	0.83 [0.43]	-1.13 [-0.61]	-0.33 [-0.18]	-2.01 [-1.05]	-1.99 [-1.04]	2.07 [1.13]
smb	-5.37 [-1.97]	-1.87 [-0.68]	-3.61 [-1.34]	-2.41 [-0.90]	-4.62 [-1.66]	-3.45 [-1.24]	-0.37 [-0.14]
hml	-5.24 [-1.42]	-8.36 [-2.19]	-2.53 [-0.70]	0.54 [0.15]	-2.66 [-0.71]	-1.83 [-0.49]	-5.39 [-1.48]
rmw	-4.95 [-1.34]	-15.22 [-3.75]	-18.75 [-4.45]	-16.95 [-4.38]	-5.83 [-1.55]	-5.84 [-1.57]	-21.61 [-4.97]
cma	5.51 [0.81]	13.61 [2.32]	12.41 [2.14]	12.25 [2.21]	25.70 [3.79]	45.28 [5.72]	18.20 [2.32]
umd	-0.77 [-0.41]	0.99 [0.53]	-1.43 [-0.77]	-2.39 [-1.30]	-0.28 [-0.15]	-0.76 [-0.40]	-3.18 [-1.77]
# months	732	728	728	728	732	732	728
$\bar{R}^2(\%)$	11	11	12	14	7	8	21

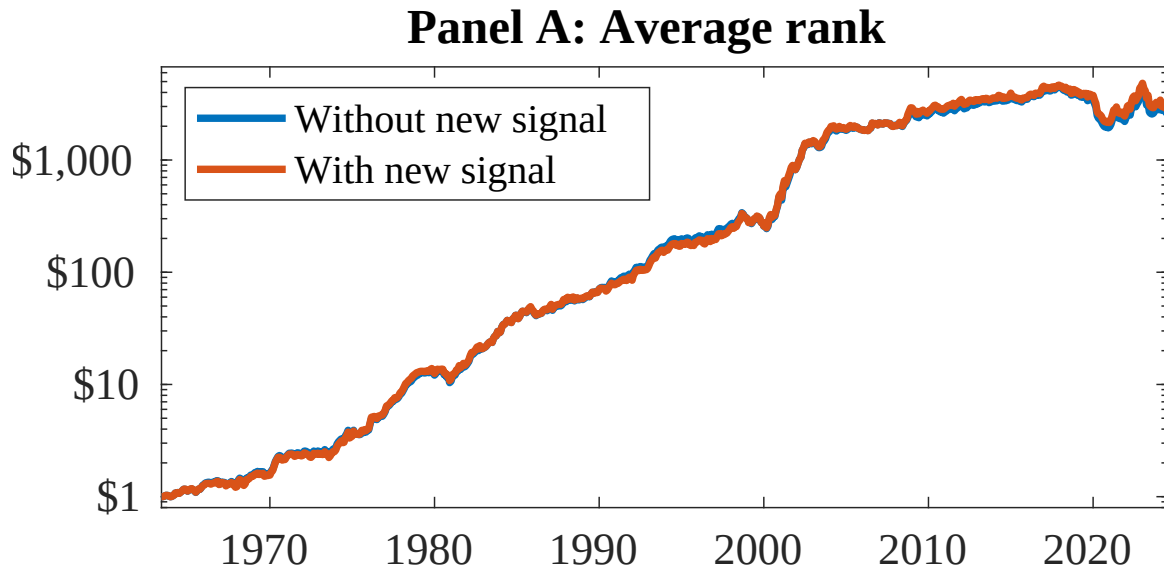


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PRS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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